

Which Regression Line Fits?

AP Statistics · Unit 5 · Topic 5.5 · B: 2027-03-09 · E: 2027-03-10 · TEACHER KEY

CED TETHER

- **Skill 2.C** — Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU DAT-1** — Regression models may allow us to predict responses to changes in an explanatory variable.
- **LO DAT-1.G** — Estimate parameters for the least-squares regression line model. [Skill 2.C]
- **EK DAT-1.G.1** — The least-squares regression model minimizes the sum of the squares of the residuals and contains the point $(\bar{x}, \bar{y})$.
- **EK DAT-1.G.2** — The slope, b , of the regression line can be calculated as $b = r(s_y / s_x)$, where r is the correlation between x and y , s_y is the sample standard deviation of the response variable, and s_x is the sample standard deviation of the explanatory variable.
- **EK DAT-1.G.3** — Sometimes, the y -intercept of the line does not have a logical interpretation in context.
- **EK DAT-1.G.4** — In simple linear regression, r^2 is the square of the correlation, r . It is also called the coefficient of determination. r^2 is the proportion of variation in the response variable that is explained by the explanatory variable in the model.
- **LO DAT-1.H** — Interpret coefficients for the least-squares regression line model. [Skill 4.B]
- **EK DAT-1.H.1** — The coefficients of the least-squares regression model are the estimated slope and y -intercept.
- **EK DAT-1.H.2** — The slope is the amount that the predicted y -value changes for every unit increase in x .
- **EK DAT-1.H.3** — The y -intercept value is the predicted value of the response variable when the explanatory variable is equal to 0. The formula for the y -intercept, a , is $a = \bar{y} - b\bar{x}$.

Essential question: How do several regression properties distinguish lines that share the same mean point?

DOK-3 skill: 4.B · **FRQ pattern:** shared-mean-point-which-line-is-lsr1

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

First take → rules + (a)–(c) → turn in

DOK: 1 → 3

Minutes: 45

Teacher does

- Show the board at the bell and collect a one-sentence first take without confirming a claim.
- Read the rules box; work (a) and (b). Allow about 10 minutes to finish (a)–(c).
- Ask for evidence linked to a claim. Collect the sheet and hand-score part (c) with E/P/I.

Students do

- Make a first claim, then use the printed formulas to work (a) and (b).
- Synthesize a supported analysis in (c), revise, and turn in.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?

- What would you tell someone who chose the other claim?

Adult role

Circulate and ask students to connect their choice to evidence. Score part (c) using the three required reasoning elements; do not require dropped extension work.

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Keeps Quinn’s line using the correct SD ratio and slope
- **reasoning-2** (required) — Explains why the shared mean point is insufficient to choose the LSRL
- **reasoning-3** (required) — Keeps Rowan’s 64 percent and interprets response variation in context

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Reversing the standard-deviation ratio
- Treating the mean point as sufficient
- Using r as explained variation

[STAR] TODAY'S PROBLEM — annotated

Suppose a solar-farm analyst uses sunshine hours x to predict daily generation y in megawatt-hours (MWh). The table summarizes 12 observed days.

Quinn: “Using $b = r(s_y/s_x)$ gives $\hat{y} = 5.6 + 3.2x$. It passes through $(7, 28)$, so it must be the LSRL. It explains 80 percent of the variation.”

Rowan: “Using $b = r(s_x/s_y)$ gives $\hat{y} = 26.6 + 0.2x$. It also passes through $(7, 28)$, so it is equally good. Explained variation is 64 percent.”

Solar-farm summary (12 observed days)

x range	$\bar{x}$	$\bar{y}$	s_x	s_y	r
4–10 h	7 h	28 MWh	1.5 h	6 MWh	0.80

FIRST TAKE (5 min)

The two lines agree at 7 sunshine hours. Does that make them equally useful at other times? Give your first thought in one sentence.

Key: Not graded. Credit a committed clause-level comparison supported by one property.

Rules callout “MAKE THE REGRESSION PROPERTIES AGREE (keep on the page)” is printed on the student sheet.

DOK 1 (a) State the mean point that every least-squares regression line must contain.

Key: The mean point is $(\bar{x}, \bar{y}) = (7, 28)$: 7 sunshine hours and 28 MWh.

DOK 2 (b) Calculate the regression slope and intercept, and r^2 . Give the slope's units.

Key: $b = 0.80(6/1.5) = 3.2$ MWh per hour; $a = 28 - 3.2(7) = 5.6$ MWh; $r^2 = (0.80)^2 = 0.64$.

DOK 3 (c) Which parts of Quinn's and Rowan's claims should a report keep? Use the slope formula to choose a line, explain why their shared mean point does not make both lines correct, and interpret the correct percent of variation explained. Write 3–4 sentences.

Key: Keep Quinn's line, $\hat{y} = 5.6 + 3.2x$, because it uses the correct response-SD over explanatory-SD ratio. Both lines pass through $(7, 28)$, but only Quinn's has the regression slope; passing through the mean point alone is not enough to minimize squared residuals. Keep Rowan's 64%: the linear model explains 64% of the variation in daily generation. Quinn's 80% confuses correlation with its square.

EXIT

One thing the rules box changed about my first take: _____